# NYC 311

# Urban Service Efficiency Analytics

SQL & Google Sheets Project



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# 1. Project Overview

For this project, I worked with NYC 311 service request data from January to March 2026. I wanted to understand what kind of problems people reported most often, which boroughs received the most requests, and how the different agencies handled them.

I used SQL in DBeaver with an SQLite database to clean and analyse the data. I focused on request volumes, agency workloads, request statuses, closure rates and resolution times. I also created a dashboard using Google Sheets to present the results and make the main findings easier to understand.

The reason I chose this project was to see how data can help identify areas that may need more attention. Instead of making assumptions, I wanted to use the information in the dataset to understand the situation better and support decisions with actual results.

# 2. Business Questions and Objectives

Before starting the analysis, I prepared some questions to understand the NYC 311 data better. I wanted to look at the number of service requests, the most common problems, and how requests were distributed across different boroughs and agencies.

**The main questions I wanted to answer were:**

1. **What are the most common problems reported by residents?**
2. **Which borough receives the highest number of service requests?**
3. **How does the number of requests change each month?**
4. **Which agencies receive the most service requests?**
5. **Which agencies receive more requests than the average?**
6. **How many requests are closed, open or still in progress?**
7. **Which problem types have the highest percentage of requests that are not closed?**
8. **Does the percentage of requests that are not closed differ between boroughs?**
9. **How long does it usually take to resolve a service request?**

The objective was to use the results to understand where most of the demand comes from, identify possible areas that need more attention, and get a better picture of how service requests are being handled.

## 3. Dataset Description

For this project, I used the NYC 311 Service Requests dataset, which I downloaded from the official NYC Open Data website. The dataset covers the period from January to March 2026 and contains 1,016,333 service request records.

The original dataset contained many columns with different types of information, such as request dates, agency details, problem descriptions, addresses and location information. However, I did not need all of them for my analysis, so I selected only the columns that were relevant to my project.

I kept information such as the request ID, creation and closure dates, agency, problem type, borough, ZIP code and request status. I also renamed the columns to make them easier to understand and use in my SQL queries.

I worked with the data using an SQLite database in DBeaver and created a separate table called clean_311_requests for my analysis. The original dataset was kept unchanged.

Data:311 Service Requests from 2020 to Present | NYC Open Data

## 4. Data Cleaning and Preparation

Before starting the analysis, I first checked the original dataset to understand its structure and identify missing values. I used SQL queries to count missing values in important columns, such as request ID, creation date, closure date, agency, problem type, status and borough.

The check showed that all records had a request ID and a creation date. I found 29,813 records without a closure date and only 31 records with a missing problem detail. The other columns included in this check did not have missing values.

I also checked the relationship between request status and missing closure dates to understand whether there were any unusual records that needed further investigation.

After checking the data, I created a new table called clean_311_requests. I selected the 10 columns needed for my analysis and renamed them to make them easier to understand and use in my queries. I formatted the creation date, removed unnecessary spaces, converted empty text values into NULL, and changed the ZIP code column to text. I kept the original dataset unchanged.

# 5. SQL Analysis

## 5.1 Most Common Problem Types

I started the analysis by looking at the most common problems reported by residents. I used SQL to count the number of service requests for each problem type and sorted the results from highest to lowest.

The results showed that Heat/Hot Water was the most reported problem, with 167,038 requests, followed by Illegal Parking with 149,261 requests. Noise - Residential was in third place, with 85,994 requests.

These results show that Heat/Hot Water and Illegal Parking were the two most common problem types in the dataset. Together, they represented around 31.1% of all service requests.

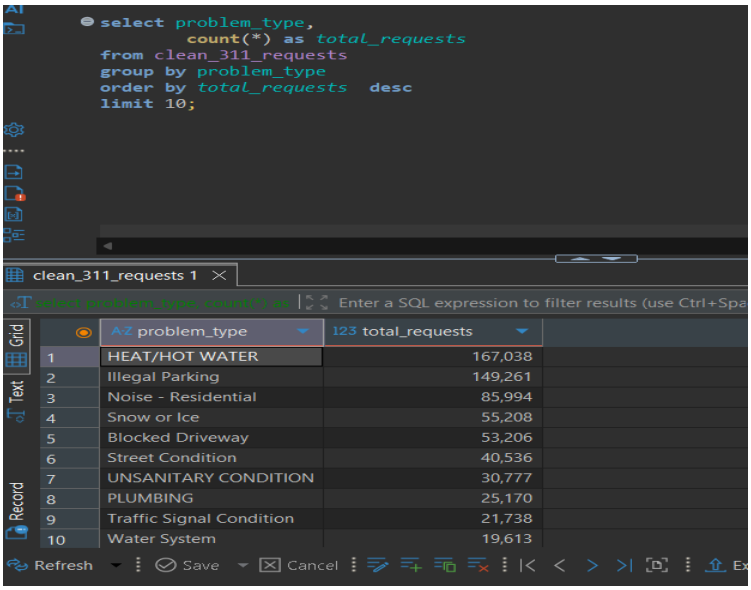



Figure 1. SQL query and results showing the 10 most common problem types.

## 5.2 Service Requests by Borough

Next, I looked at how service requests were distributed across the five boroughs of New York City. I used SQL to count the number of requests for each borough and calculate their percentage of the total.

Brooklyn had the highest number of requests, with 310,999, representing 30.60% of all requests. Queens came second with 246,591 requests (24.26%), followed by the Bronx with 214,839 (21.14%). Manhattan had 193,042 requests (18.99%), while Staten Island had the lowest number, with 50,038 requests (4.92%).

The results show that Brooklyn and Queens together accounted for around 54.9% of all service requests in the dataset.

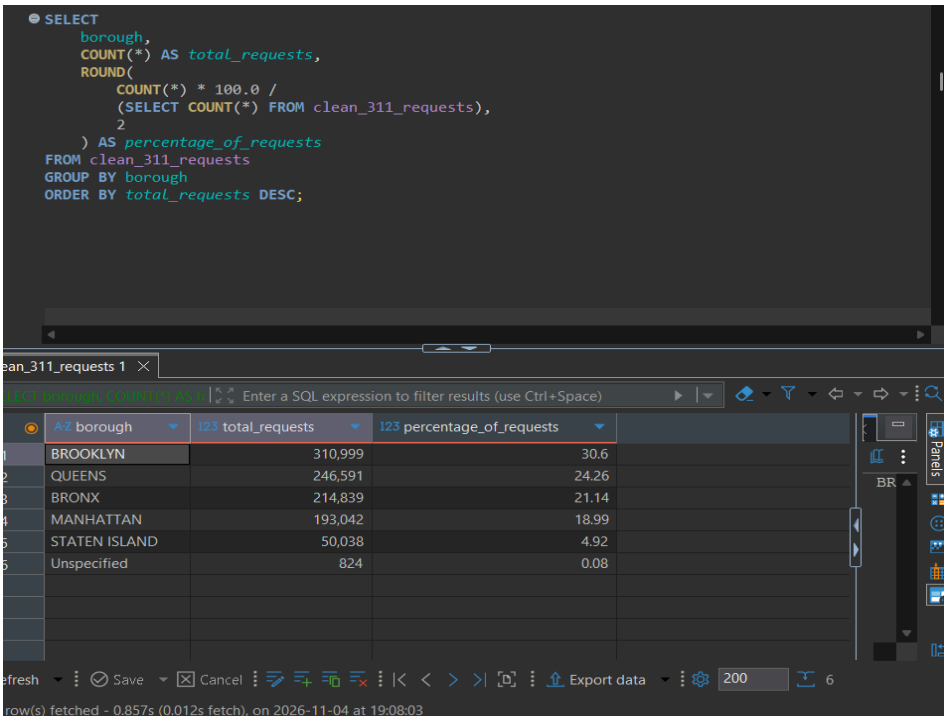



Figure 2. SQL query and results showing Service Requests by Borough and percentage.

## 5.3 Monthly Service Request Trends

I also wanted to understand how the number of service requests changed over the three months. I used SQL to group the requests by month using their creation dates and count how many requests were received each month.

January had the highest number of requests, with 342,123. In February, the number decreased to 334,691, before increasing again to 339,519 in March.

Overall, the number of service requests remained fairly stable during the three months, with only small changes from one month to another.

## 5.4 Request Status Distribution

I then looked at the status of the service requests to understand how many had been closed and how many were still being handled.

The results showed that 982,733 requests were marked as Closed, which represents approximately 96.7% of all requests. There were also 14,458 requests marked as Open and 12,381 marked as In Progress. The remaining requests had statuses such as Assigned, Pending and Started.

This shows that most requests in the dataset were marked as closed. However, the status alone does not tell us how quickly or how well each request was resolved.

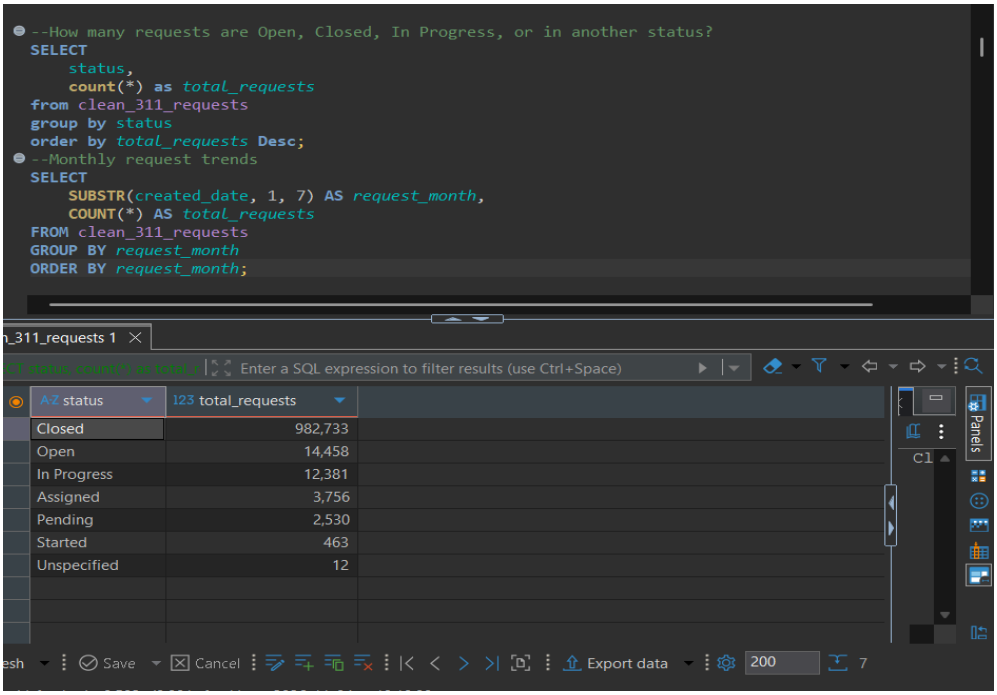



Fig 3. Request Status Distribution.

## 5.5 Closure Rates by Problem Type

After looking at the most common problem types, I wanted to compare their closure rates to see how many requests had been marked as closed.

The results showed that Illegal Parking and Blocked Driveway both had a closure rate of 100%. Snow or Ice also had a very high rate of 99.79%. On the other hand, Traffic Signal Condition had the lowest closure rate among the top 10 problem types, at 88.99%, followed by Street Condition at 92.70% and Heat/Hot Water at 94.41%.

This comparison shows that closure rates were different depending on the type of problem. However, these percentages only show the request status at the time covered by the dataset. They do not tell us how quickly the requests were handled or how satisfied residents were with the outcome.

## 5.6 Service Requests by Agency

I also analysed how service requests were distributed across the different agencies. I used SQL to count the number of requests assigned to each agency and compare their workloads.

The results showed that NYPD received the highest number of requests, with 367,427, followed by HPD with 309,052. DSNY was in third place with 116,419 requests, followed by DOT with 86,632.

NYPD and HPD together accounted for around 66.56% of all service requests. This shows that a large part of the requests in the dataset was assigned to these two agencies.

However, the number of requests alone does not show how well an agency performed, as different agencies deal with different types of problems.

## 5.7 Agencies with Above-Average Request Volume

After comparing the number of requests received by each agency, I wanted to identify which agencies had a higher workload than the average.

I used a SQL subquery to calculate the average number of requests across all 16 agencies. The average was approximately 63,521 requests per agency.

The results showed that four agencies were above this average: NYPD, HPD, DSNY and DOT.

This analysis helped me compare the workload across agencies and identify which ones received more requests than the overall average. However, a higher number of requests does not necessarily mean that an agency performs better or worse than another.

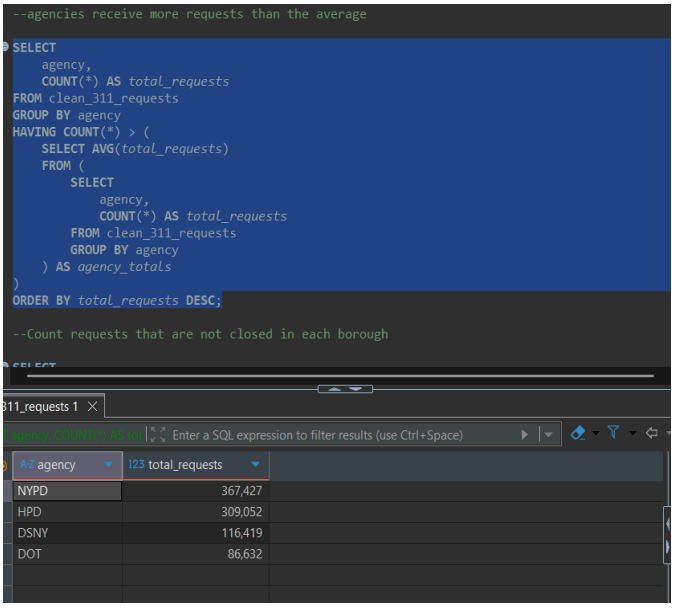



Fig 4. Agencies have requests more than Average.

## 5.8 Problem Types with the Highest Percentage of Not-Closed Requests

I also looked at which problem types had the highest percentage of requests that were not marked as Closed. This helped me identify some less common problems that had a high number of unresolved requests compared with their total number of requests.

One interesting result was Noise - Helicopter. All 3,289 requests in this category were still marked as In Progress, meaning that none had been marked as Closed in the dataset. I found a similar result for Construction Lead Dust, where all 350 requests were also marked as In Progress.

These results caught my attention because both problem types had a 100% not-closed rate. However, this does not necessarily mean that the requests were never resolved. It only shows their recorded status in the dataset during the period analysed.

## 5.9 Requests Not Closed by Borough

I also wanted to compare the number of requests that were not marked as Closed across the different boroughs. I used SQL to count these requests and calculate their percentage for each borough.

The results showed that Manhattan had the highest percentage of requests not closed, at 4.92%, followed by the Bronx at 3.59% and Queens at 3.30%. Brooklyn had a rate of 2.37%, while Staten Island had the lowest rate, at 1.55%.

Although Manhattan had the highest percentage, Brooklyn had the highest total number of requests overall. This shows why it is useful to compare percentages as well as total numbers.

These results show some differences between boroughs, but they do not not necessarily mean that one borough receives better service than another. The types of problems reported and other factors may also affect these results.

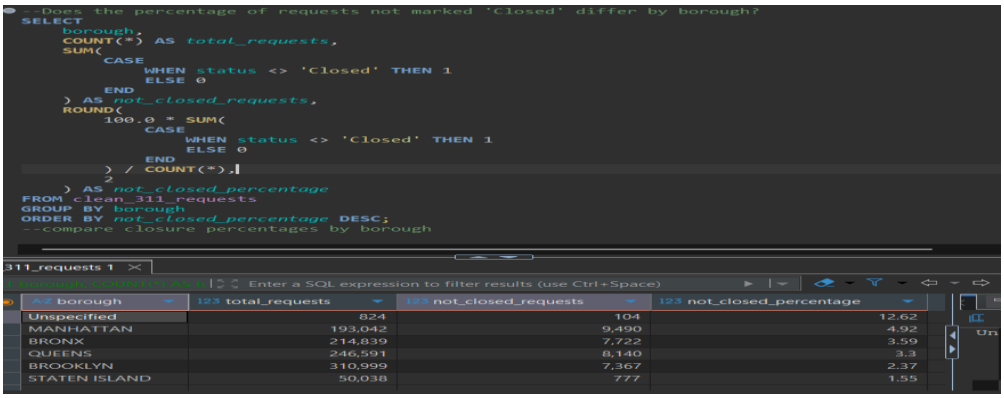



Fig 5. Request not closed by borough.

## 5.10 Average Resolution Time

Finally, I looked at how long it took to close service requests. I used SQL to calculate the time between the creation date and the closure date for requests with valid, non-negative resolution times.

To do this, I first converted the date values into a format that SQLite could use for calculations. Then, I calculated the difference between the creation date and closure date in hours and used the AVG function to find the average resolution time.

The results showed that the average resolution time was approximately 190.46 hours, which is equal to 7.94 days.

This gives an idea of how long it took to close requests on average during the period analysed. However, the average can be affected by cases that take much longer to resolve, so it should be interpreted with care.

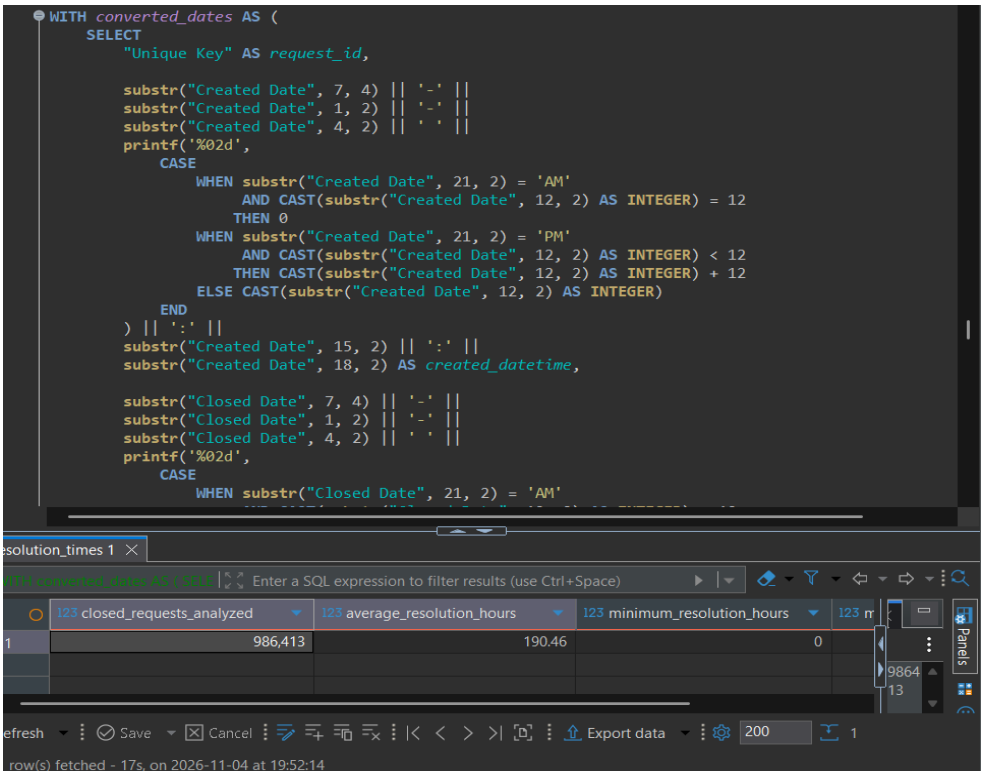



Fig 6. Average Resolution Time

## 6. Dashboard and Data Visualisation

After completing the SQL analysis, I used Google Sheets to create a dashboard to present the results in a more visual and easier-to-understand way.

I created four KPI cards to show the main figures of the project: total service requests, total agencies, closed requests and average resolution time.

I also created nine charts to explore different parts of the data. These include service requests by borough, monthly request trends, request status distribution, the most common problem types and the number of requests received by each agency. I also included charts to compare closure percentages and requests that were not closed across different problem types and boroughs.

I chose different chart types depending on the information I wanted to present. For example, I used bar charts to compare request volumes, a pie chart to show the share of requests by borough, and a column chart to show monthly trends.

The dashboard brings the main results together on one page, making it easier to compare the different categories and understand the main patterns in the data.

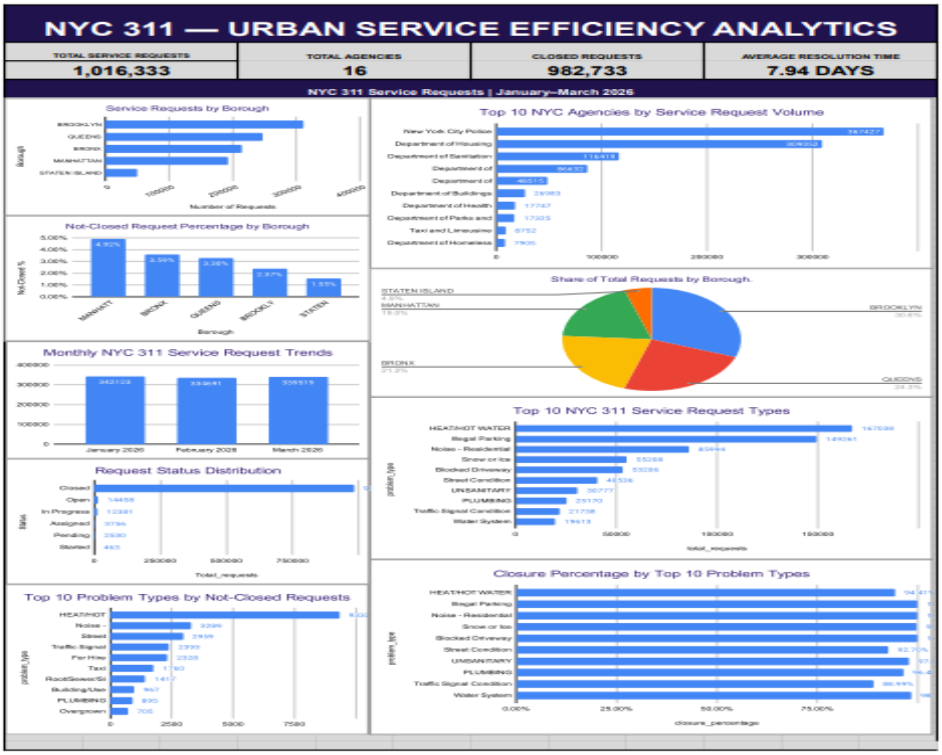



Fig 7. Dashboard

## 7. Key Findings and Conclusion

After completing the analysis, I found several interesting patterns in the NYC 311 service request data for January to March 2026.

One of the main findings was that Heat/Hot Water was the most common problem reported by residents, with 167,038 requests, followed by Illegal Parking with 149,261 requests.

Brooklyn had the highest number of service requests, with 310,999, representing 30.60% of all requests. Queens came second with 246,591 requests. Together, these two boroughs accounted for around 54.9% of all service requests.

The analysis also showed that NYPD and HPD received the highest number of requests, with 367,427 and 309,052 respectively. Together, they accounted for approximately 66.56% of all requests.

Another interesting finding was that 982,733 requests were marked as Closed, representing approximately 96.7% of the total. However, some problem types had a much higher percentage of requests that were not closed. For example, all 3,289 Noise - Helicopter requests were still marked as In Progress in the dataset.

Finally, I calculated the average resolution time for requests with valid resolution times. The result was approximately 7.94 days.

Overall, this project helped me understand how SQL can be used to clean, organise and analyse a large dataset. It also showed me how a dashboard can make the results easier to understand and compare. I learned how to use different SQL techniques, including aggregations, JOINs and subqueries, and how to present the findings using Google Sheets.

## 8. Limitations and Future Improvements

Although this project helped me understand the NYC 311 service request data, there are some limitations to consider.

First, the analysis only covers three months, from January to March 2026. A longer period would make it possible to identify seasonal patterns and compare changes over time.

Another limitation is that the analysis is mainly based on the number and status of service requests. These figures do not necessarily show the quality of the service provided by each agency or how satisfied residents were with the results.

The average resolution time also needs to be interpreted carefully because it only includes requests with valid resolution times and recorded closure dates. In addition, a small number of requests that took much longer to resolve could affect the average.

If I continued this project, I would like to analyse a longer period of data, compare results across different years and investigate resolution times in more detail. I would also like to explore whether certain problem types take longer to resolve in different boroughs.

These improvements could provide a more detailed understanding of service requests and help identify patterns that are not visible in a three-month analysis.